AI Hype vs Investment

Does Interest Track Where the Money Is Going?

A Data Science Project Analyzing AI Public Interest vs. Capital Allocation

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Twelve Economies · Four Public Sources · Zero Invented Numbers

Table of Contents

AI Hype vs Investment: Does Interest Track Where the Money Is Going?

Public interest in AI and capital committed to AI, on the surface, looks like two readings of the same thing. They aren't. This is a data science project — twelve economies, four public sources, no invented numbers — built to find out how far apart they actually are, and whether five years closes the gap.

Every figure below traces to a named report: the World Bank (GDP, population), Google Trends (search interest), Stanford HAI's AI Index (private investment, read from the primary chapter PDFs, not a summary), and the IMF's AI Preparedness Index. Twelve countries survived that data-availability discipline, down from an original plan for fifteen.

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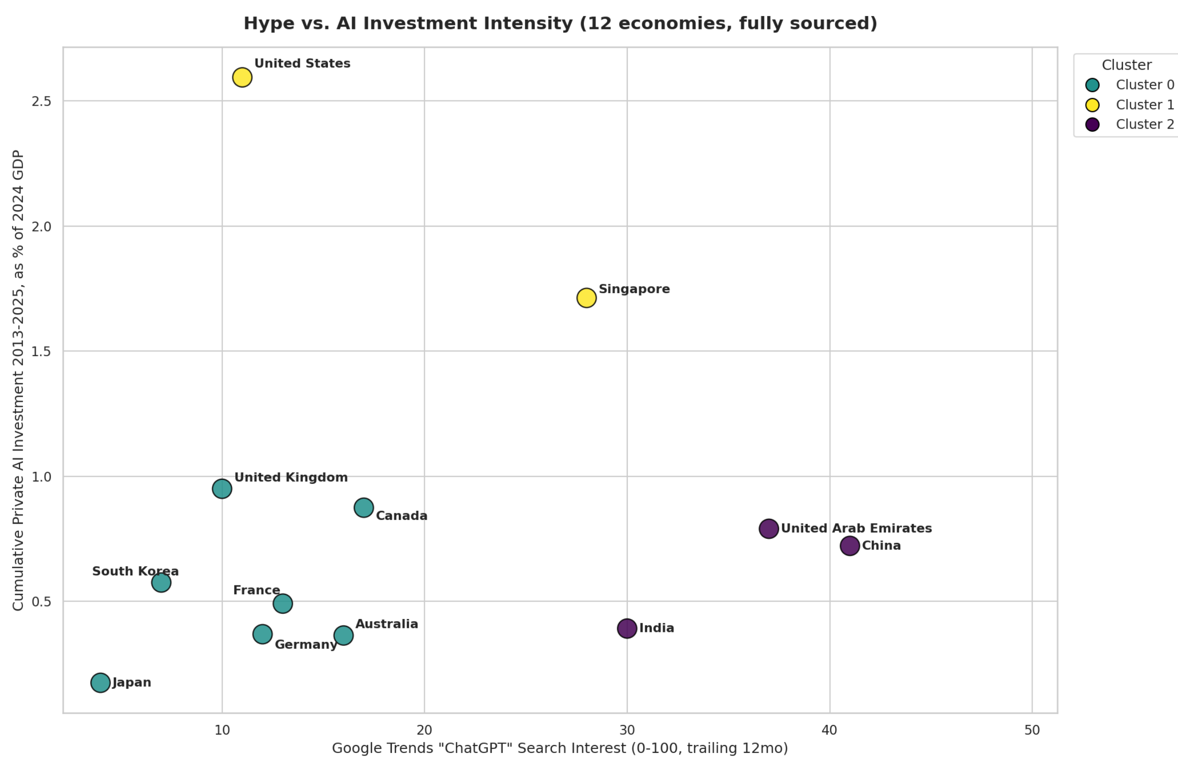
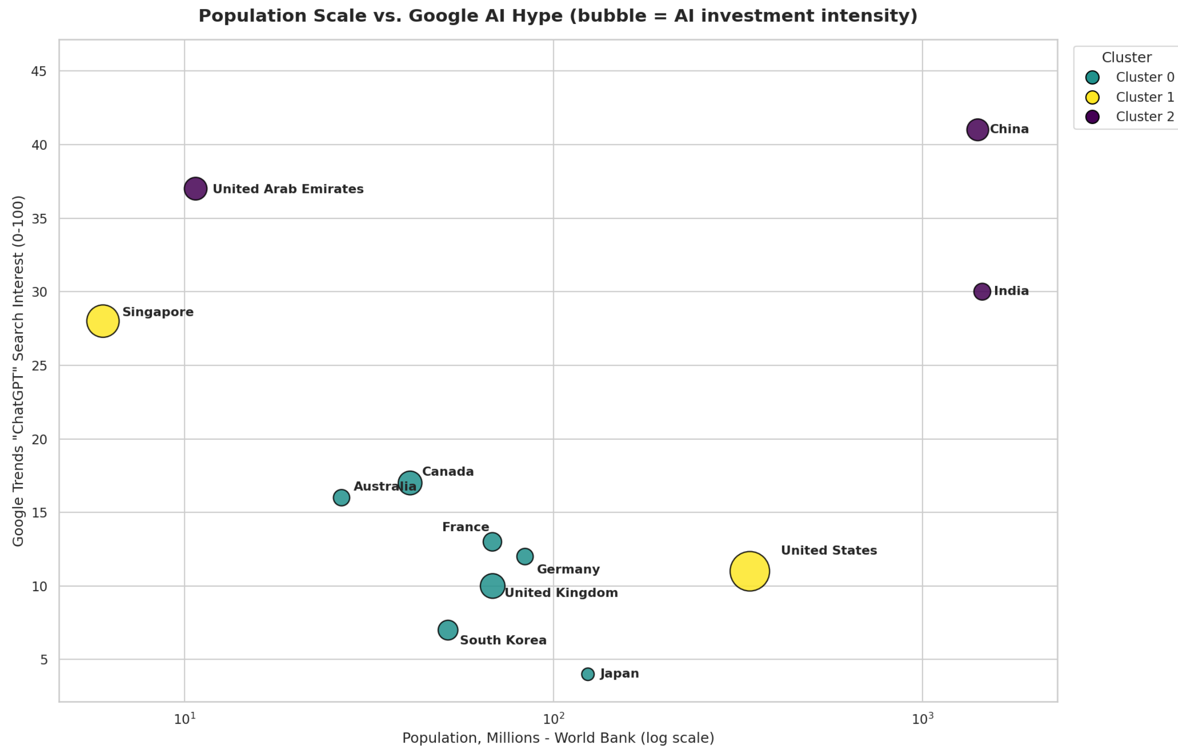
1. Where hype sits against investment, currently



Hype vs. AI investment intensity, 12 economies, fully sourced

The United States posts one of the lowest AI search-interest scores of the twelve — behind the UK, Germany, even South Korea — and by far the highest investment intensity, at 2.6% of GDP. Singapore's high hype and above-average investment intensity reflect a real capability. China and the UAE post the two highest hype scores in the set, with India close behind and sitting mid-pack on capital.

2. Scale doesn't explain it either



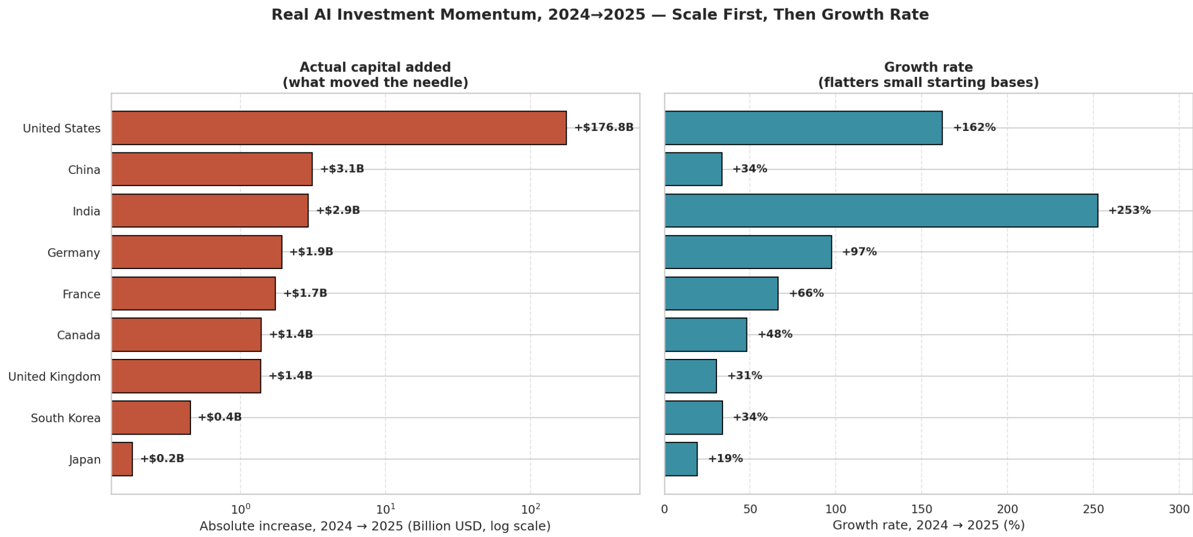
Population scale vs. Google AI hype; bubble size = investment intensity

China and India's high hype scores aren't just a population effect. Google Trends doesn't count total searches — it measures what share of each country's own search activity was about AI, on a 0–100 scale. So China's score of 41 means AI makes up a large slice of China's search attention, not that China simply has more people typing "ChatGPT" or "Claude" into Google.

What the US & Singapore bubble shows here is simply how much capital concentration doesn't come with matching public curiosity. Two different playbooks get to very different outcomes here. The US leans on deep capital markets, tech giants, and a dense research ecosystem, while Singapore leans on fast, centralized policy coordination and targeted public-private funding.

The UAE, with Singapore, makes the point even sharper: with populations of 11M and 6M, they post hype scores of 37 and 28 — in the same range as China and India, whose populations are each over 200 times larger. Population isn't driving this metric in either direction.

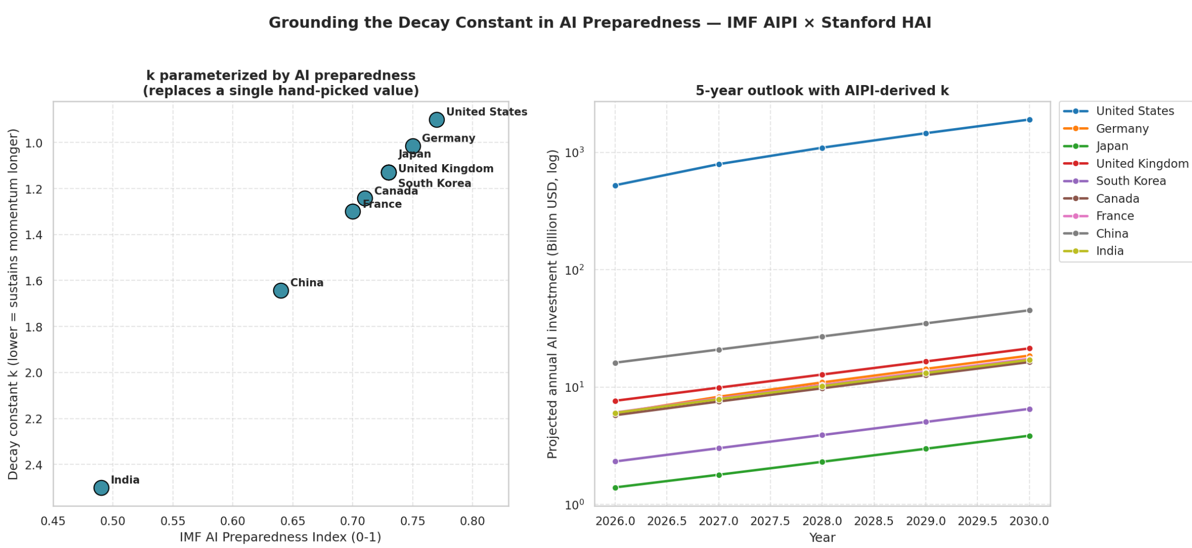
3. A number everyone will misread if you let them



Real 2024→2025 AI investment momentum — growth rate vs. actual capital added

India's private AI investment grew 253% in 2025 — the fastest of any country here, faster than the US. Sorted by growth rate, it looks like the story. Sorted by actual capital added, it's $2.9B — less than China's $3.1B at a "sluggish" 34%, and 1.6% of what the US added in the same year. Both numbers are true. Showing only the percentage is how you get an inflated take on LinkedIn.

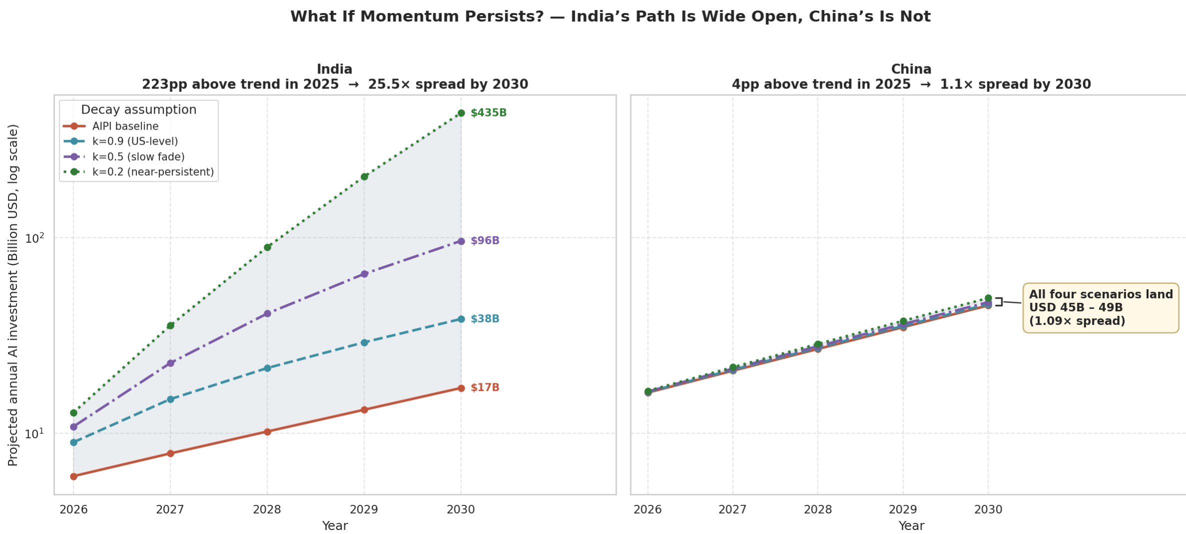
4. A forecast needs an assumption — so make the assumption defensible



Decay constant grounded in the IMF AI Preparedness Index, vs. a hand-picked value

Projecting investment forward means assuming how fast an unusual year of growth fades back to trend. Rather than pick that number by feel, it's ordered by the IMF's AI Preparedness Index: more-prepared economies are assumed to sustain above-trend growth longer. The chart shows what that produces for nine countries with a real 2024→2025 growth observation to build from.

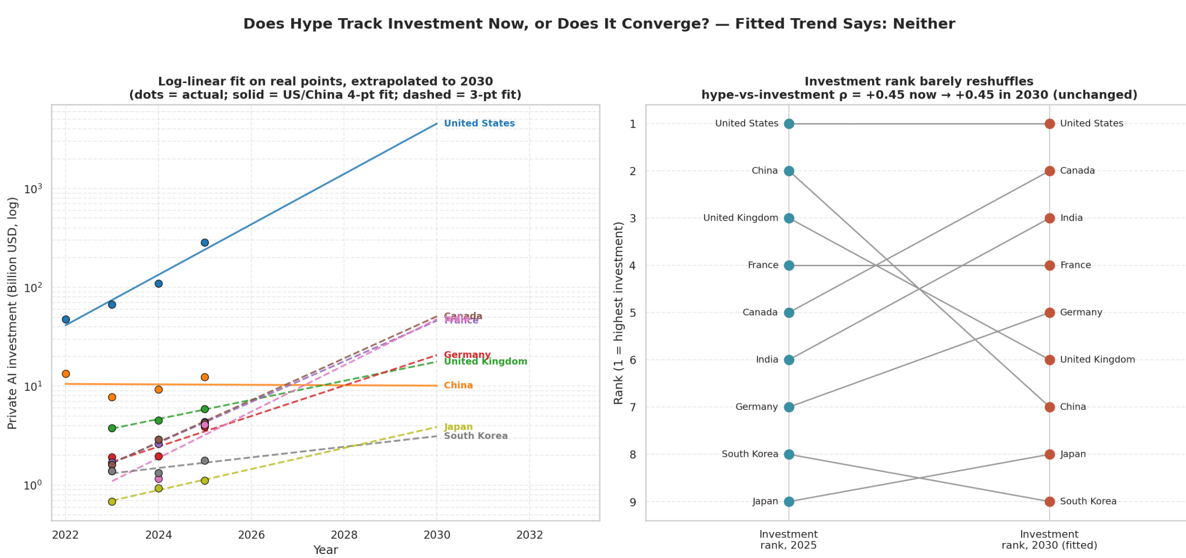
5. Where the assumption matters — and where it doesn't at all



Scenario fan: India vs. China under four different decay assumptions

This is the honest test of that assumption. China's 2025 growth sat just 4 points above its long-run trend — so every decay scenario, fast or slow, lands within a 1.09× range. Its future is close to locked in regardless of the model. India's growth sat 223 points above trend, so the same range of assumptions produces a 25.5× spread — anywhere from $17B to $435B by 2030. Whether India's jump was a structural shift (multi-year corporate commitments already booked) or a one-off deal-timing spike is the single largest open question in this dataset, and it's a question about intent and capacity that no decay model can settle by itself.

6. So, does hype converge with investment, or not?



Log-linear fit on real points, and hype-vs-investment rank correlation, now vs. 2030

Fitting a real trend line to each country's actual reported figures (not the hand-tuned decay model) and ranking hype against investment both today and under the 2030 extrapolation gives a rank correlation of $\rho = +0.45$ in both cases — unchanged. Under this model, hype and capital hold the same loose relationship in five years that they hold today. It doesn't tighten. It doesn't fall apart. It just persists, imperfectly, the whole way through.

What this actually says

Hype is a leading indicator of attention, not of capital deployment — and the data gives no evidence that gap is closing on its own. A few structural reasons help explain why:

State capital vs. market capital. China's $12.4B private-investment figure and its own government's ~$177B domestic AI industry estimate aren't in conflict — Stanford HAI's methodology is private-VC only, and structurally excludes the state guidance funds that drive much of China's AI spending. Comparing the US and China on a single "investment" number quietly assumes they're organized the same way. They aren't.

A blind spot in the hype metric itself. China's low Google Trends score partly reflects a market where the tools being searched for are inaccessible — real engagement happens inside domestic apps this metric doesn't see. The same logic runs the other way for Gemini everywhere: pre-installed on Android, embedded in Search, generating real usage that never becomes a "search."

The US as exporter, not searcher. Low domestic hype fits a country that builds and exports frontier models rather than discovers or consumes them.

Small, centralized, and fast. The UAE and Singapore's high hype and above-average investment intensity reflect a real capability — centralized governments in small, wealthy states can mandate AI adoption at a pace larger economies can't match structurally, not just statistically.

What isn't in this dataset

Investment is one input. Compute capacity, chip supply chains, energy availability, and regulatory posture all shape whether committed capital becomes deployed capability — none of that is modeled here, and it shouldn't be asserted without the same sourcing discipline as everything above. That's the natural next layer, built the same way this one was: find the primary source, or leave the country out.

Country theses: instinct vs. data

Three reads on this dataset, from three different starting points — public sentiment absorbed over time, then checked against what was actually measured. Each is broken into what the numbers confirm and what the thesis adds beyond them, because those are different kinds of claims and shouldn't be presented as if they carry equal weight.

India — volume without absorption

The thesis: a huge AI "consumer" base, brain drain, compute scarcity, underinvested R&D, and a services-and-outsourcing culture — so hype runs far ahead of investment, and stays there.

What the data confirms: India posts the second-highest hype score in the set (30, behind only China's 41) — consistent with a large, active individual-user base. It simultaneously posts the lowest AI Preparedness Index of the nine countries with a full profile (0.49, against a US of 0.77) and sits at the bottom on investment intensity (0.39% of GDP, tied with Germany and Australia for last). High hype, low structural readiness, low capital — three independently sourced numbers, and they line up exactly the way the thesis predicts.

What goes beyond the data: why AIPI is low. The thesis names several specific mechanisms — compute scarcity, brain drain, an outsourcing-oriented culture. AIPI is a composite score — this project has the total, not the sub-scores that would let these individual causes be told apart from, say, human capital and re-skilling, or data-center infrastructure. The direction is confirmed; the named causes are a plausible read, not something isolated in what's measured here.

China — closed-loop monetization

The thesis: local, niche innovation — developed and consumed domestically — is what swings investment upward, in a way outside trackers can't easily see.

What the data confirms: this is close to a direct match for something already sourced two separate ways. Stanford HAI's private-investment figure for China is $12.4B — small next to the US, and Stanford's own report explicitly notes this understates China's real AI spending, because it structurally excludes the state guidance funds that drive much of it. China's own government (CAICT/MIIT) puts domestic AI industry size closer to $177B for 2025. A roughly 14x gap between "what a Western VC tracker sees" and "what China's own numbers say," on the same underlying activity — that's the closed-loop dynamic the thesis describes.

What goes beyond the data: "niche" as a description of the innovation itself. The CAICT figure confirms scale that Western tracking misses — it doesn't characterize what kind of innovation is happening domestically (niche/applied vs. broad/frontier) but few initiatives like "smart factory" stand out as proof of extreme academic scale. That part is a reasonable read of a closed AI ecosystem, not something either sourced number actually measures.

United States — capacity over curiosity

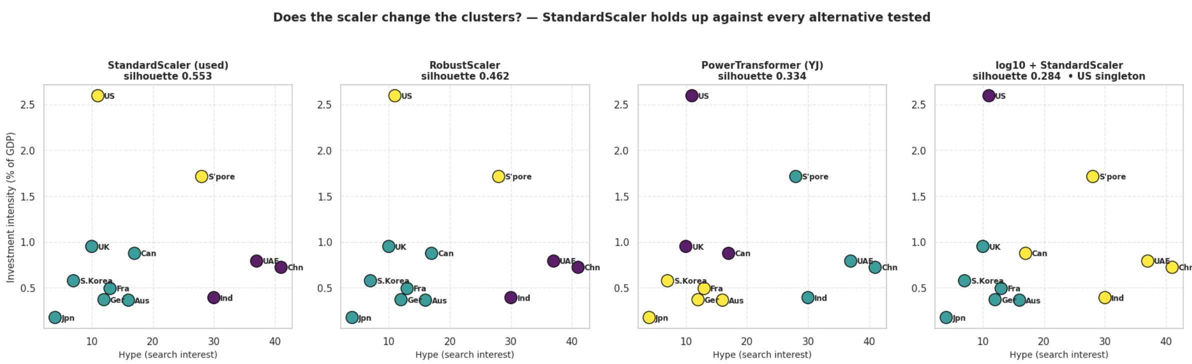
The thesis: wins the race outright, with moderate-to-low public hype, because it's occupied with building frontier models and infrastructure for the world to consume rather than generating domestic search interest.

What the data confirms: this is the cleanest match of the three, on numbers already central to the whole project. Lowest hype score in the set (11 — below the UK, Germany, even South Korea) paired with the highest investment intensity by a wide margin (2.6% of GDP, roughly 1.5x the next-closest country). That's exactly "capacity-focused, not curiosity-focused" in two independently sourced metrics.

What goes beyond the data: "wins the race outright." Investment intensity and low domestic hype are confirmed; whether that constitutes "winning" depends on how the race is defined — capital deployed, frontier capability, deployed-model market share, and long-run economic return aren't the same finish line, and this project only measures the first two, not the others.

Appendix: does the scaler change the clusters?

StandardScaler is outlier-sensitive in theory — this dataset has one (the US at 2.60% investment intensity vs. a 0.55% median). Rather than assume that's a problem, it was tested against three alternatives: RobustScaler (built specifically to resist outliers), PowerTransformer, and a log10 transform. The chart below shows all four, side by side, on the same real 12-country data.



KMeans clustering under four different scalers — StandardScaler, RobustScaler, PowerTransformer, log10 + StandardScaler

The result: it isn't a problem. RobustScaler — outlier-immune by construction — returns identical cluster membership to StandardScaler (100% agreement), and StandardScaler scores highest on silhouette (0.553) of every variant tested, including RobustScaler (0.462), PowerTransformer (0.335), and log10 + StandardScaler (0.284). The one variant that meaningfully differs is log10 + StandardScaler, which isolates the US as its own singleton cluster — arguably more interpretable, since the US genuinely is a category of one, but it scores worse on separation. A DBSCAN check (outlier-native by design, rather than forced into k clusters) flags the US, Japan, Singapore, and India as noise on the log-scaled data — a different way of saying the same thing: a handful of countries don't sit comfortably in any one group, and StandardScaler's three-cluster split is a reasonable summary of that, not an artifact of the chosen transform.

Sources

World Bank Open Data (GDP, population) · Google Trends via pytrends · Stanford HAI AI Index 2025 & 2026, Chapter 4 (Economy) · IMF AI Preparedness Index (2023, 174 economies) · China AI industry size — CAICT/MIIT via China Daily/Xinhua reporting.

Methodology notes: *built using StandardScaler + KMeans clustering and a log-linear/exponential-decay forecast model, benchmarked in the full notebook against RobustScaler, PowerTransformer, and QuantileTransformer alternatives (StandardScaler held up against all of them), and against a properly-scaled Huber regression check. Full notebook, data, and sourcing on GitHub — every number is traceable to a citation, and every assumption is labeled as one.*